

Row Echelon form:-

Diagonal of a matrix:- (Leading, Principal, Primary, Main, major diagonal)

In any matrix $[a_{ij}]$, the entries like a_{ii} i.e. entries at i^{th} Row, i^{th} column are called diagonal entries

For example:-

$$\begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \\ 11 & 12 & 13 & 14 & 15 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \\ 13 & 14 & 15 \end{bmatrix}$$

Row Echelon form:-

Any matrix is of the form that, the ~~below~~ entries below the leading diagonal are zero, is called Row Echelon form

For example:-

$$\begin{bmatrix} 2 & 3 & 4 \\ 0 & 6 & 7 \\ 0 & 0 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 7 & 8 & 9 & 10 \\ 0 & 0 & 13 & 14 & 15 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 6 \\ 0 & 0 & 9 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Row Elementary operation (Row-Preserving operation)

Any matrix can be reduced to Row Echelon form by Row elementary operations, they are

- (i) Interchanging of any 2-rows ($R_1 \leftrightarrow R_2$)
- (ii) Multiplication of every element of ~~any~~ row by any non-zero scalar. ($R_2 \rightarrow \lambda R_2$, where $\lambda \neq 0$)
- (iii) Adding a scalar multiplied Row to another Row ($R_3 \rightarrow R_3 + \lambda R_2$)

Rank of a matrix :-

Minor of a matrix A

Determinant of any sub-square matrix of a given matrix A is called minor of A. If the sub-square matrix is of order γ then minor is of order γ .

For ex:-

$$A = \begin{bmatrix} \textcircled{2} & 3 & \textcircled{4} \\ 5 & \textcircled{6} & 7 \\ \textcircled{1} & 2 & \textcircled{5} \end{bmatrix}_{3 \times 3}$$

$B = \begin{bmatrix} 2 & 4 \\ 1 & 5 \end{bmatrix}$ is a sub-square matrix of order 2.
 $|B| = 10 - 4 = 6$ is a minor of A of order 2.

$C = \begin{bmatrix} 6 & 7 \\ 2 & 5 \end{bmatrix}$ is also a sub-square matrix of order 2.

$|C| = 30 - 14 = 16$ is a minor of A of order 2.

$|A| = -6$ is also minor of A of order 3.

Since A itself sub-square matrix of A.

Rank of a matrix :-

Let A be a $m \times n$ matrix. The rank of A is said to be ' γ ' if (i) there is at least one minor of A of order γ which does not vanish. and

(ii) Every minor of A of order $(\gamma+1)$ & higher order vanishes.

In other words,

The rank of a matrix is the largest of the orders of all the non-vanishing minors of that matrix.

Procedure to find Rank of a matrix:-

Step 1: Reduce the given matrix into Row Echelon Form.

Step 2: Rank of A = No of non-zero rows in the Row Echelon form.

System of Linear Algebraic Equations:-

Consider the following system of m linear equations in n unknowns.

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

⋮

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

The matrix Representation of above system be

$$\boxed{AX = B}$$

where $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$ be the coefficient matrix of the system

$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ be the matrix of unknowns & $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$ be the matrix of constants.

System of Homogeneous Linear Equations & Non-Homo Linear Equations:-

If $B \equiv O$, a zero matrix, then the system $AX = O$ is called system of Homogeneous Linear Equations.

otherwise, the system $AX = B$ is called non-Homogeneous Linear Equations.

Augmented matrix:-

Appending column matrix B to the coefficient matrix A as the additional last column, is called Augmented matrix of the system. & it is denoted by

$$[A|B] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right]$$

Solution of the system of linear equations:-

A set of values of $x_1, x_2, \dots, x_n$ which satisfy all the given m equations in the system simultaneously is called a solution of the system.

Consistent & Inconsistent System :-

If the system of equation has a solution, then it is consistent otherwise, ~~it is~~ the system is inconsistent.

Note:- A consistent system may have either only one (or) infinitely many solution, when the system has only one solution, it is called the unique solution.

System of Non-Homogeneous Linear Equations:-

Rouché's Theorem (Nec & Suf condition of a system of linear non-homogeneous equations is to be consistent)

The system of equations $AX = B$ is consistent, if and only if

the coefficient matrix A & the augmented matrix $[A|B]$ are of the same rank. $[R(A) = R(A|B)]$

To discuss the consistency of the system in detail:-

- (i) If $R(A) \neq R(A|B)$, the equations are inconsistent.
- (ii) If $R(A) = R(A|B) = n$, the no of unknowns, then the equations are consistent & have a unique solution.
- (iii) If $R(A) = R(A|B) < n$, the no of unknowns, then the equations are consistent & have an infinite number of solutions.

System of Homogeneous Linear Equations:-

3.

The System of Homogeneous Linear Equations

$$A X = 0 \quad (m \text{ equations in } n \text{ unknowns})$$

This system is always consistent, since by Rouche's Theorem

$$R(A) = R(A|0)$$

Trivial Solution:-

If the coeff matrix A is non-singular (i.e. $|A| \neq 0$), then the system has a unique solution, namely zero solution

$$(i.e. \ x_1 = x_2 = x_3 = \dots = x_n = 0)$$

This unique solution is called the trivial solution.
(which is not of any importance)

Non-Trivial Solution:-

If the coeff matrix A is singular (i.e. $|A| = 0$), then the system has an infinite number of non-zero (or) non-trivial solutions.

1) Test the consistency of the following system of equations & solve if consistent.

$$\begin{aligned} (i) \quad & 3x + y + z = 8 \\ & -x + y - 2z = -5 \\ & x + y + z = 6 \\ & -2x + 2y - 3z = -7 \end{aligned}$$

Soln:- STEP 1:-

The matrix Representation of the system is

$$\begin{bmatrix} 3 & 1 & 1 \\ -1 & 1 & -2 \\ 1 & 1 & 1 \\ -2 & 1 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \\ 6 \\ -7 \end{bmatrix}$$

Step 2:- Rank of Augmented matrix

$$[A|B] = \left[\begin{array}{ccc|c} 3 & 1 & 1 & 8 \\ -1 & 1 & -2 & -5 \\ 1 & 1 & 1 & 6 \\ -2 & 2 & -3 & -7 \end{array} \right] \quad R_1 \leftrightarrow R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ -1 & 1 & -2 & -5 \\ 3 & 1 & 1 & 8 \\ -2 & 2 & -3 & -7 \end{array} \right] \quad \begin{array}{l} R_2 \rightarrow R_2 + R_1 \\ R_3 \rightarrow R_3 - 3R_1 \\ R_4 \rightarrow R_4 + 2R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 2 & -1 & 1 \\ 0 & -2 & -2 & -10 \\ 0 & 4 & -1 & 5 \end{array} \right] \quad \begin{array}{l} R_3 \rightarrow R_3 + R_2 \\ R_4 \rightarrow R_4 - 2R_2 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & -3 & -9 \\ 0 & 0 & 1 & 3 \end{array} \right] \quad R_4 \rightarrow 3R_4 + R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & -3 & -9 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \text{which is Echelon form of } [A|B].$$

Now $R(A|B) = \text{No of Non-zero Rows in the Echelon form}$
 $= 3.$

$$\text{Also } R(A) = 3$$

Hence By Rouché's Theorem

$$R(A|B) = R(A) = 3, \text{ No of unknowns.}$$

$\therefore$ The system is consistent & has a unique solution.

Steps Solution of the System...

4.

From step 2, we identified equivalent matrix of $[A|B]$

$$[A|B] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & -3 & -9 \end{array} \right]$$

From the above, we get.

$$x + y + z = 6$$

$$2y - z = 1$$

$$-3z = -9 \Rightarrow z = 3$$

$$x = 6 - 2 - 3 = 1$$

$$2y = 4 \Rightarrow y = 2$$

Solving above by back substitution, we get

$$\boxed{x=1, y=2, z=3}$$

(ii)

$$\begin{aligned} 2x_1 + x_2 + 2x_3 + x_4 &= 6 \\ x_1 - x_2 + x_3 + 2x_4 &= 6 \\ 4x_1 + 3x_2 + 3x_3 - 3x_4 &= -1 \\ 2x_1 + 2x_2 - x_3 + x_4 &= 10 \end{aligned}$$

Sol: - Step 1: The matrix Representation of the system is

$$\begin{bmatrix} 2 & 1 & 2 & 1 \\ 1 & -1 & 1 & 2 \\ 4 & 3 & 3 & -3 \\ 2 & 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \\ -1 \\ 10 \end{bmatrix}$$

Step 2: Rank of $[A|B]$ & $[A]$

$$[A|B] = \left[\begin{array}{cccc|c} 2 & 1 & 2 & 1 & 6 \\ 1 & -1 & 1 & 2 & 6 \\ 4 & 3 & 3 & -3 & -1 \\ 2 & 2 & -1 & 1 & 10 \end{array} \right] R_1 \leftrightarrow R_2$$

$$[A|B] \sim \left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 6 \\ 2 & 1 & 2 & 1 & 6 \\ 4 & 3 & 3 & -3 & -1 \\ 2 & 2 & -1 & 1 & 10 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 4R_1 \\ R_4 \rightarrow R_4 - 2R_1 \end{array}$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 6 \\ 0 & 3 & 0 & -3 & -6 \\ 0 & 7 & -1 & -11 & -25 \\ 0 & 4 & -3 & -3 & -2 \end{array} \right] R_2 \rightarrow \frac{R_2}{3}$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 6 \\ 0 & 1 & 0 & -1 & -2 \\ 0 & 7 & -1 & -11 & -25 \\ 0 & 4 & -3 & -3 & -2 \end{array} \right] \begin{array}{l} R_3 \rightarrow R_3 - 7R_2 \\ R_4 \rightarrow R_4 - 4R_2 \end{array}$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 6 \\ 0 & 1 & 0 & -1 & -2 \\ 0 & 0 & -1 & -4 & -11 \\ 0 & 0 & -3 & 1 & 6 \end{array} \right] R_4 \rightarrow R_4 + 3R_3$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 6 \\ 0 & 1 & 0 & -1 & -2 \\ 0 & 0 & -1 & -4 & -11 \\ 0 & 0 & 0 & 13 & 39 \end{array} \right] \text{which is row Echelon form}$$

Now

$$R(A|B) = \text{No of Non-Zero Rows in E.F.} \\ = 4$$

Also $R(A) = 4$.

Hence by Rouche's Theorem.

$$R(A|B) = R(A) = 4, \text{ No of unknowns}$$

$\therefore$ The System is consistent & has a unique solution.

Step 3:- Solution of the system:-

5.

From Echelon form of step 2, we get

$$\begin{array}{rcl} x_1 - x_2 + x_3 + 2x_4 & = & 6 \\ x_2 - x_4 & = & -2 \\ -x_3 - 4x_4 & = & -11 \\ 13x_4 & = & 39 \end{array} \quad \left| \quad \begin{array}{l} x_1 = 6 + 1 + 1 - 6 = 2 \\ x_2 = -2 + 3 = 1 \\ x_3 = -4(3) + 11 = -1 \\ \Rightarrow x_4 = 3 \end{array} \right.$$

Solve the above by back substitution, we get

$$\boxed{x_1 = 2, x_2 = 1, x_3 = -1, x_4 = 3}$$

(iii) $2x + y + 5z + w = -10$

$$x + y + 3z - 4w = -1$$

$$3x + 6y - 2z + w = 8$$

$$2x + 2y + 2z - 3w = 2$$

Sol:- Step 1:- The matrix representation of the above system

$$\begin{bmatrix} 2 & 1 & 5 & 1 \\ 1 & 1 & 3 & -4 \\ 3 & 6 & -2 & 1 \\ 2 & 2 & 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} -10 \\ -1 \\ 8 \\ 2 \end{bmatrix}$$

Step 2 Rank of Augmented matrix

$$[A|B] = \left[\begin{array}{cccc|c} 2 & 1 & 5 & 1 & -10 \\ 1 & 1 & 3 & -4 & -1 \\ 3 & 6 & -2 & 1 & 8 \\ 2 & 2 & 2 & -3 & 2 \end{array} \right] \quad R_2 \leftrightarrow R_1$$

$$[A|B] \sim \left[\begin{array}{cccc|c} 1 & 1 & 3 & -4 & -1 \\ 2 & 1 & 5 & 1 & 5 \\ 3 & 6 & -2 & 1 & 8 \\ 2 & 2 & 2 & -3 & 2 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \\ R_4 \rightarrow R_4 - 2R_1 \end{array}$$

$$\sim \left[\begin{array}{cccc|c} 1 & 1 & 3 & -4 & -1 \\ 0 & -1 & -1 & 9 & 7 \\ 0 & 3 & -11 & 13 & 11 \\ 0 & 0 & -4 & 5 & 4 \end{array} \right] R_3 \rightarrow R_3 + 3R_2$$

$$\sim \left[\begin{array}{cccc|c} 1 & 1 & 3 & -4 & -1 \\ 0 & -1 & -1 & 9 & 7 \\ 0 & 0 & -14 & 40 & 32 \\ 0 & 0 & -4 & 5 & 4 \end{array} \right] R_3 \rightarrow \frac{R_3}{-14}$$

$$\sim \left[\begin{array}{cccc|c} 1 & 1 & 3 & -4 & -1 \\ 0 & -1 & -1 & 9 & 7 \\ 0 & 0 & 1 & \frac{-20}{7} & \frac{-16}{7} \\ 0 & 0 & -4 & 5 & 4 \end{array} \right] R_4 \rightarrow R_4 + 4R_3$$

$$\sim \left[\begin{array}{cccc|c} 1 & 1 & 3 & -4 & -1 \\ 0 & -1 & -1 & 9 & 7 \\ 0 & 0 & 1 & \frac{-20}{7} & \frac{-16}{7} \\ 0 & 0 & 0 & \frac{-45}{7} & \frac{-36}{7} \end{array} \right] \text{which is req Echelon form}$$

Now $R[A|B] = \text{No of Non-Zero Row}$
 $= 4.$

$$\times R[A] = 4.$$

Hence by Rouché's Thm, $R[A|B] = R[A] = 4$, No of unknown

$\therefore$ The system is consistent & has a unique solution.

Step 3: Solution of the System 1.

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From Echelon form of step 2, we set

$$x + 2y + 3z - 4w = -1$$

$$-y - z + 9w = 7$$

$$z - \frac{20}{7}w = -\frac{16}{7}$$

$$-\frac{45}{7}w = -\frac{36}{7}$$

$$\Rightarrow x = -1 - \frac{1}{7} + 4\left(\frac{4}{5}\right) = 2.$$

$$\Rightarrow y = 0 + 9\left(\frac{4}{5}\right) - 7 = \frac{1}{5}$$

$$\Rightarrow z = \frac{-16}{7} + \frac{20}{7}\left(\frac{4}{5}\right) = 0$$

$$\Rightarrow w = \frac{36}{45} = \frac{4}{5}$$

Solve the above by back substitution, we get

$$\boxed{x=2, y=\frac{1}{5}, z=0, w=\frac{4}{5}}$$

② Test the consistency of the following system of equations & solve if consistent

$$(i) \quad x - 4y - 3z + 16 = 0$$

$$2x + 7y + 12z = 48$$

$$4x - y + 6z = 16$$

$$5x - 5y + 3z = 0$$

Soln.

Step 1: The matrix rep is given by

$$\begin{bmatrix} 1 & -4 & -3 \\ 2 & 7 & 12 \\ 4 & -1 & 6 \\ 5 & -5 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -16 \\ 48 \\ 16 \\ 0 \end{bmatrix}$$

Step 2 The Rank of Augmented matrix:

$$[A|B] \sim \left[\begin{array}{ccc|c} 1 & -4 & -3 & -16 \\ 2 & 7 & 12 & 48 \\ 4 & -1 & 6 & 16 \\ 5 & -5 & 3 & 0 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 4R_1 \\ R_4 \rightarrow R_4 - 5R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -4 & -3 & -16 \\ 0 & 15 & 18 & 80 \\ 0 & 15 & 18 & 80 \\ 0 & 15 & 18 & 80 \end{array} \right] \begin{array}{l} R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - R_2 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -4 & -3 & -16 \\ 0 & 15 & 18 & 80 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \text{ which is the req Echelon form}$$

Now $R[A|B] = \text{No of Non-Zero Row}$
 $= 2$

* $R[A] = 2$

Hence by Rouché's Theorem, $R(A|B) = R(A) < 3$, No of unknown

$\therefore$ The system is consistent, but it has infinite NO of solution.

Step 3: Family of Solution of the System:

From the Echelon form of Step 2, we get

~~$x_1 + 4x_2 + 3x_3 = -16$~~

$$x - 4y - 3z = -16$$

$$15y + 18z = 80$$

$$x = 4\left(\frac{16}{3} - \frac{6}{5}k\right) + 3k - 16$$

$$15y = 80 - 18k$$

$$y = \frac{80}{15} - \frac{18}{15}k = \frac{16}{3} - \frac{6}{5}k$$



Let $z = k$, where k is a Parameter (Free)

Hence we get,

$$x = \frac{16}{3} - \frac{9k}{5}, \quad y = \frac{16}{3} - \frac{6}{5}k, \quad z = k$$

where k is a Parameter
which are one Parameter family of Solutions

$$\begin{aligned} x &= \frac{64}{3} - \frac{24}{5}k + 3k - 16 \\ &= \frac{64 - 48}{3} + \frac{-24k + 15k}{5} \\ &= \frac{16}{3} + \frac{-9k}{5} \\ &= \frac{16}{3} - \frac{9k}{5} \end{aligned}$$

Note: $\left. \begin{array}{l} \text{No. of Parameters} \\ \text{in a Family of Solution} \end{array} \right\} = \text{Difference between No. of unknowns \& Common rank of A \& [A|B].}$

(i) $x_1 + 2x_2 - x_3 - 5x_4 = 4$

$x_1 + 3x_2 - 2x_3 - 7x_4 = 5$

$2x_1 - x_2 + 3x_3 = 3$

Sol: - Step 1: The matrix rep of above is

$$\begin{bmatrix} 1 & 2 & -1 & -5 \\ 1 & 3 & -2 & -7 \\ 2 & -1 & 3 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix}$$

Step 2: Rank of Augmented matrix.

$$[A|B] = \left[\begin{array}{cccc|c} 1 & 2 & -1 & -5 & 4 \\ 1 & 3 & -2 & -7 & 5 \\ 2 & -1 & 3 & 0 & 3 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array}$$

$$\sim \left[\begin{array}{cccc|c} 1 & 2 & -1 & -5 & 4 \\ 0 & 1 & -1 & -2 & 1 \\ 0 & -5 & 5 & 10 & -5 \end{array} \right] R_3 \rightarrow R_3 + 5R_2$$

$$\sim \left[\begin{array}{cccc|c} 1 & 2 & -1 & -5 & 4 \\ 0 & 1 & -1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Now $R[A|2] = \underline{\text{No}}$ of Non-Zero Rows
 $= 2$

$\therefore R[A] = 2$

hence by Rouché's Theorem, $R[A|B] = R[A] < 4$, No of unknowns,
 $\therefore$ The system is consistent, but it has infinite no of solutions.

Step 3:- Solution family of the system:-

From the Echelon form of Step 2, we get

$$\begin{aligned} x_1 + 2x_2 - x_3 - 5x_4 &= 4 \\ x_2 - x_3 - 2x_4 &= 1 \end{aligned} \quad \begin{aligned} \uparrow x_1 &= -2(k_1 + 2k_2 + 1) + k_1 + 5k_2 + 4 \\ x_2 &= k_1 + 2k_2 + 1 \end{aligned}$$

Let $x_3 = k_1$, $x_4 = k_2$, where k_1, k_2 are parameters.

Hence we get,

~~$x_1 = -3k_1 + 9k_2 + 6$~~

$\begin{aligned} x_1 &= -k_1 + k_2 + 2 \\ x_2 &= k_1 + 2k_2 + 1 \end{aligned}$
$\begin{aligned} x_3 &= k_1 \\ x_4 &= k_2 \end{aligned}$

which are 2-parameter family of solutions.

∴) Test the consistency of the following system of equations.

(i) $x - 2y - 3z = 2$

$3x - 2y = -1$

$-2y - 3z = 2$

$y + 2z = 1$

Step 1: The matrix eqn is given by

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$$\begin{bmatrix} 1 & -2 & -3 \\ 3 & -2 & 0 \\ 0 & -2 & -3 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 2 \\ 1 \end{bmatrix}$$

Step 2: The Rank of Augmented matrix.

$$[A|B] = \left[\begin{array}{ccc|c} 1 & -2 & -3 & 2 \\ 3 & -2 & 0 & -1 \\ 0 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{array} \right] R_2 \rightarrow R_2 - 3R_1$$

$$\begin{array}{ccc|c} 3 & -2 & 0 & -1 \\ -3 & +6 & +9 & -6 \\ \hline 0 & 4 & 9 & -7 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & -3 & 2 \\ 0 & 4 & 9 & -7 \\ 0 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{array} \right] \begin{array}{l} R_3 \rightarrow 2R_3 + R_2 \\ R_4 \rightarrow 4R_4 - R_2 \end{array}$$

$$\begin{array}{ccc|c} 0 & -4 & -6 & 4 \\ 0 & 4 & 9 & -7 \\ \hline 0 & 0 & 3 & -3 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & -3 & 2 \\ 0 & 4 & 9 & -7 \\ 0 & 0 & 3 & -3 \\ 0 & 0 & -1 & 11 \end{array} \right] R_4 \rightarrow 3R_4 + R_3$$

$$\begin{array}{ccc|c} 0 & 4 & 8 & 4 \\ 0 & -4 & -9 & +7 \\ \hline 0 & 0 & -1 & 11 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & -3 & 2 \\ 0 & 4 & 9 & -7 \\ 0 & 0 & 3 & -3 \\ 0 & 0 & 0 & 30 \end{array} \right]$$

which is the req Echelon form

Now $R[A|B] = \underline{\text{No}}$ of Non-Zero row.
= 4

$$\& R[A] = 3.$$

Hence by Rouché's Theorem $R[A|B] \neq R[A]$

$\therefore$ The system is inconsistent.

$\&$ Hence it has No. Solution

————— x —————

$$\begin{aligned}
 (ii) \quad & x_1 + x_2 + x_3 = 1 \\
 & 2x_1 - 2x_2 + 3x_3 = 1 \\
 & x_1 - x_2 + 2x_3 = 5 \\
 & x_1 + x_2 + x_3 = 2
 \end{aligned}$$

Sol:- The matrix eq is given by

Step 1:-

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & -2 & 3 \\ 1 & -1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 5 \\ 2 \end{bmatrix}$$

Step 2:- ~~The~~ Rank of the Augmented matrix :-

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 2 & -2 & 3 & 1 \\ 1 & -1 & 2 & 5 \\ 1 & 1 & 1 & 2 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1 \\ R_4 \rightarrow R_4 - R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -4 & 1 & -1 \\ 0 & -2 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{array} \right] R_3 \rightarrow 2R_3 - R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -4 & 1 & -1 \\ 0 & 0 & 1 & 9 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

which is the req Echelon form

$$\text{Now } R[A|B] = \underline{\text{No}} \text{ of non-Zero Rows} = 4$$

$$\& R[A] = 3$$

Hence by Ronche's Theorem, $R[A|B] \neq R[A]$

$\therefore$ The system is inconsistent & it has no solution

④ (i) Find the values of k for which the equations $x+y+z=1$,
 $x+2y+4z=k$ & $x+4y+10z=k^2$ may have a solution.
 For these values of k , find the solution also.

Solⁿ:

Given that, $x+y+z=1$.

$$x+2y+4z=k$$

$$x+4y+10z=k^2.$$

Step 1:

The matrix representation of above system is given by

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ k \\ k^2 \end{bmatrix}$$

Step 2: Rank of Augmented matrix:

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & k \\ 1 & 4 & 10 & k^2 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & k-1 \\ 0 & 3 & 9 & k^2-1 \end{array} \right] R_3 \rightarrow R_3 - 3R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & k-1 \\ 0 & 0 & 0 & k^2-3k+2 \end{array} \right] \text{ which is the req Echelon form.}$$

Here, we infer that, $R[A] = 3$